

EXPERIMENT 12

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Question 1:

Question:

Design a trigger such that whenever there is an insertion on the student table, the currently inserted or deleted row should be printed as it is in the output console window.

Answer:

```
CREATE OR REPLACE FUNCTION fn_student_audit()
```

```
RETURNS TRIGGER
```

```
LANGUAGE plpgsql
```

```
AS $$
```

```
BEGIN
```

```
    IF TG_OP = 'INSERT' THEN
```

```
        RAISE NOTICE 'Inserted Row -> ID: %, Name: %, Age: %, Class: %',  
NEW.id, NEW.name, NEW.age, NEW.class;
```

```
        RETURN NEW;
```

```
    ELSIF TG_OP = 'DELETE' THEN
```

```
        RAISE NOTICE 'Deleted Row -> ID: %, Name: %, Age: %, Class: %',  
OLD.id, OLD.name, OLD.age, OLD.class;
```

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RETURN OLD;

END IF;

RETURN NULL;

END;

\$\$;

Question 2:

Question:

Design a PostgreSQL trigger such that:

- Whenever a new employee is inserted into tbl_employee, a record should be added to tbl_employee_audit with the message:
"Employee name <emp_name> has been added at <current_time>"
- Whenever an employee is deleted from tbl_employee, a record should be added to tbl_employee_audit with the message:
"Employee name <emp_name> has been deleted at <current_time>"

The solution must use PostgreSQL triggers.

Answer:

```
CREATE TABLE tbl_employee (  emp_id
SERIAL PRIMARY KEY,      emp_name
VARCHAR(100) NOT NULL,
emp_salary INT
);
```

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```
CREATE TABLE tbl_employee_audit (  
sno    SERIAL    PRIMARY    KEY,  
message TEXT  
);  
  
CREATE OR REPLACE FUNCTION audit_employee_changes()  
RETURNS TRIGGER  
LANGUAGE plpgsql  
AS $$  
BEGIN  
    IF TG_OP = 'INSERT' THEN  
        INSERT INTO tbl_employee_audit(message)  
        VALUES (  
            'Employee name ' || NEW.emp_name ||  
            ' has been added at ' || NOW()  
        );  
  
        RETURN NEW;  
  
    ELSIF TG_OP = 'DELETE' THEN  
        INSERT INTO tbl_employee_audit(message)  
        VALUES (  
            'Employee name ' || OLD.emp_name ||  
            ' has been deleted at ' || NOW()  
        );
```

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```
RETURN OLD;  
END IF;
```

```
RETURN NULL;  
END;  
$$;
```

```
CREATE TRIGGER trg_employee_audit  
AFTER INSERT OR DELETE  
ON tbl_employee  
FOR EACH ROW  
EXECUTE FUNCTION audit_employee_changes();
```

Testing:

```
-- Insert an employee  
INSERT INTO tbl_employee(emp_name, emp_salary)  
VALUES ('Aman', 50000);  
  
-- Delete an employee  
DELETE FROM tbl_employee  
WHERE emp_name = 'Aman';  
  
-- Check audit log  
SELECT * FROM tbl_employee_audit;
```

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```
57 INSERT INTO tbl_employee(emp_name, emp_salary) VALUES ('Aman', 50000);
58
59 -- Delete an employee
60 DELETE FROM tbl_employee WHERE emp_name = 'Aman';
61
62 -- Check audit log
63 SELECT * FROM tbl_employee_audit;
```

Data Output Messages Notifications

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sno [PK] integer	message text
1	Employee name Aman has been added at 2026-09-22 14:20:19.212888+05...
2	Employee name Aman has been deleted at 2026-09-22 14:20:26.359906+05...